Notes: Fuster, Lo, and Willen (JF, 2024)
The Time-Varying Price of Financial Intermediation in the Mortgage Market

Contents

1 Big picture

- **Question.** How much do mortgage intermediaries charge for connecting borrowers to MBS investors, and how does this price vary with demand, monetary policy, and postcrisis regulation?
- **Object of interest.** The price of intermediation, ϕ, is the gap between the secondary-market value of the loan and the primary-market payment to the borrower through rebates.
- **Measurement innovation.** The paper combines proprietary Optimal Blue lender rate sheets with MBS/TBA prices to measure daily and monthly intermediation prices directly.
- **Main results.** Pass-through from MBS prices to rebates is strong but incomplete; high application volume reduces pass-through; ϕ rose substantially from 2008 to 2014.
- **Aggregate implication.** Total spending on intermediation over October 2008 to October 2014 is estimated at about \$155 billion; the upward trend added roughly \$97 billion in borrower costs.

2 Data and measurement

The core data are proprietary Optimal Blue rate sheets for conforming fixed-rate mortgages. The baseline loan is \$300,000, 80% LTV, FICO 750, 30-year fixed, owner-occupied, no prepayment penalty, 30-day lock, Los Angeles, California. The main rate-sheet sample contains 343,133 lender–day–rate observations from 61 lenders over 1,523 days.

Interpretation. The rate sheet is the primary-market price menu faced by loan officers and borrowers. It lets the authors measure what borrowers receive for locking a mortgage with a given note rate.

The secondary-market value uses TBA prices for agency MBS, adjusted for guarantee fees and loan-level price adjustments. The gain-on-sale measure π adjusts for the net cost of servicing by using mortgage servicing-right valuation multiples. The difference $\phi - \pi$ is interpreted as the present value of net servicing costs.

Interpretation. The paper separates the broad borrower-paid intermediation wedge from origination markups and servicing costs.

3 Models and empirical specifications

3.1 Price of intermediation

$$p^n_{YSP} = 100 + YSP(r_n), \qquad \phi^n \equiv p^n_{TBA} - p^n_{YSP}.$$

Interpretation. The intermediary buys the loan from the borrower by paying principal plus rebate and sells the economic exposure in the MBS market. The wedge ϕ is the implicit price paid for origination, underwriting, delivery, and servicing.

3.2 Intermediary supply

$$\max_{q_n} \ \phi^n q_n - C(q_n), \qquad C'(q_n) = \phi^n.$$

Interpretation. If marginal cost is increasing, an MBS price increase shifts demand outward, but part of the gain raises ϕ instead of borrower rebates. This is the core incomplete pass-through mechanism.

3.3 Origination, servicing, and gain-on-sale

$$C'(q_n) = C'_o(q_n) + PV(c_s),$$
$$\pi^n = p_{TBA}^{n-0.25} + 0.25 \times mult_{MIAC}^n - p_{YSP}^n.$$

Interpretation. π measures what the intermediary earns from origination after valuing the mortgage servicing right. The difference $\phi - \pi$ measures the net servicing component.

3.4 Pass-through and monthly time-variation regressions

$$\Delta p_{YSP,t} = \alpha + \beta \Delta p_{TBA,t} + \varepsilon_t,$$
$$\phi_{l,t} = \alpha_l + \beta Applications_t + \tau Trend_t + \Gamma X_t + \varepsilon_{l,t}.$$

Interpretation. β measures how much a one-dollar MBS price change moves borrower rebates. The monthly regressions quantify the relation between demand, trend, volatility, concentration, and the price components.

4 Identification and empirical design

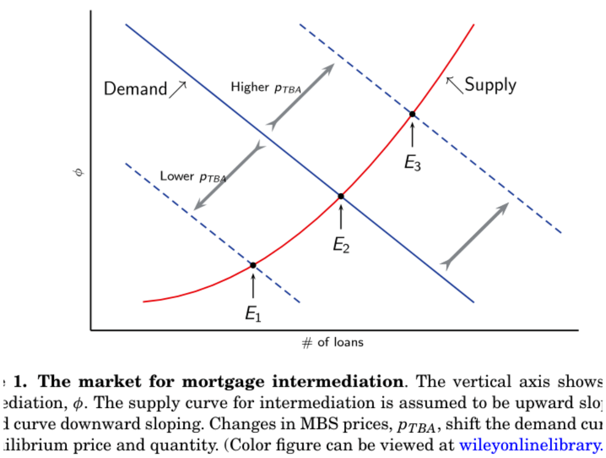
The paper combines QE event studies, daily pass-through regressions, monthly lender regressions, IV specifications using WAC minus 10-year Treasury yield, processing-time tests, MSA-level regressions, and MBA cost data. The design is not a randomized experiment; credibility comes from the consistency of evidence across high-frequency prices, demand proxies, capacity measures, local tests, and independent cost data.

5 Tables and figures with detailed interpretation

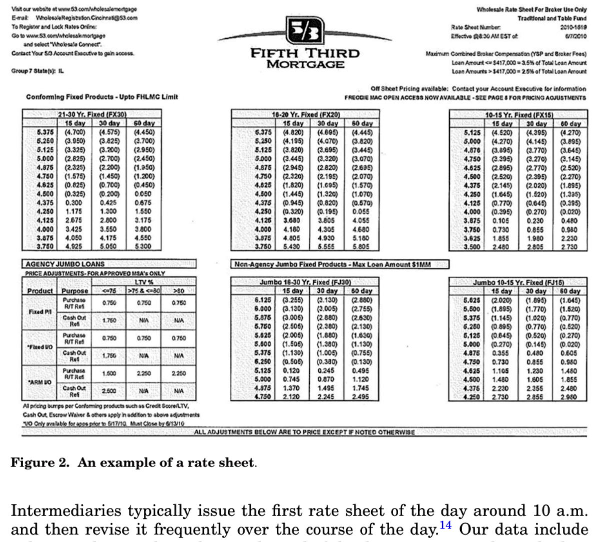
5.1 Figure 1: The market for mortgage intermediation and Figure 2: An example of a rate sheet

Read: Figure 1 shows an upward-sloping supply curve for intermediation and a demand curve shifted by MBS/TBA prices. Figure 2 shows the raw lender rate-sheet menu of rates and rebates.

Economic message. The conceptual model and raw pricing menu together explain the measurement strategy: compare secondary-market loan values to borrower rebates to infer the intermediary wedge.



(a) Figure 1: The market for mortgage intermediation



(b) Figure 2: An example of a rate sheet

5.2 Table I: Characteristics of the proprietary lender rate sheet data from Optimal Blue and Figure 3: Mortgage rates over our sample period

Read: Table I reports 21.72 lenders per day, 10.37 offers per lender-day, mean rate 4.53%, mean rebate 1.08, and mean rebate/rate slope 4.58. Figure 3 shows Rate101 falling during the crisis and rising again after 2012.

Economic message. The data are rich enough to measure the price menu facing borrowers and to map secondary-market MBS price changes into primary-market rebates.

Table I

Characteristics of the Proprietary Lender Rate Sheet Data from Optimal Blue

Panel A shows assumed values for various loan characteristics underlying the offers we use for the baseline scenario in our analysis. Panel B shows various descriptive statistics of the rate sheet offers for loans with these exact characteristics (where an offer is a combination of note rate and points). Definitions are as follows: for "daily mean across lenders," we compute the mean for each day and then take moments across days. For "within-lender spread," we calculate the maximum minus minimum offer for each lender-day. For "daily 90-10 spread across lenders," we take lender-day observations and then take the spread between the 90th and 10th percentiles for each day. The "rebate/rate slope" is the slope of an OLS regression of rebates on rates computed for each lender-day. "Rate101" is constructed for each lender-day by interpolating between different offers to obtain a note rate that yields a rebate of +1 point (or $p_{YSP} = 101$). The sample in Panel B comprises 343,133 offers from 62 unique lenders across 1,523 days. The start date is October 16, 2008, and the end date is October 31, 2014.

Panel A: Loan-Level Characteristics for Baseline Scenario

Variable	Value	Variable	Value
Loan amount	300,000	Prepayment penalty	None
Loan-to-value ratio (LTV)	80	Lock period	30 days
Credit score (FICO)	750	Owner-occupied or investment	Owner-occupied
Program	Conforming	State	CA
Loan type	Fixed	MSA	Los Angeles
Term	30 year		

Panel B: Descriptive Statistics of Rate Sheet Data

			Percentile				
	Mean	*SD*	10	25	50	75	90
Lenders per day	21.72	5.55	13	18	21	27	28
Offers per lender-day	10.37	2.92	7	8	10	12	15
Rate offered							
Daily mean across lenders	4.53	0.55	3.73	4.19	4.48	4.93	5.18
Within-lender spread	1.17	0.16	1	1	1.13	1.25	1.38
Rebate offered							
Daily mean across lenders	1.08	0.61	0.27	0.56	1.16	1.58	1.82
Within-lender spread	5.26	1.08	3.64	4.82	5.27	6.00	6.62
Rebate/Rate slope							
Daily mean across lenders	4.58	1.20	3.00	3.96	4.61	5.62	6.02
Daily 90-10 spread across lenders	1.71	0.70	0.89	1.15	1.62	2.11	2.69
Rate101							
Daily mean across lenders	4.41	0.66	3.50	3.94	4.37	4.86	5.21
Daily 90-10 spread across lenders	0.29	0.15	0.18	0.21	0.25	0.31	0.42

(a) Table I: Optimal Blue data

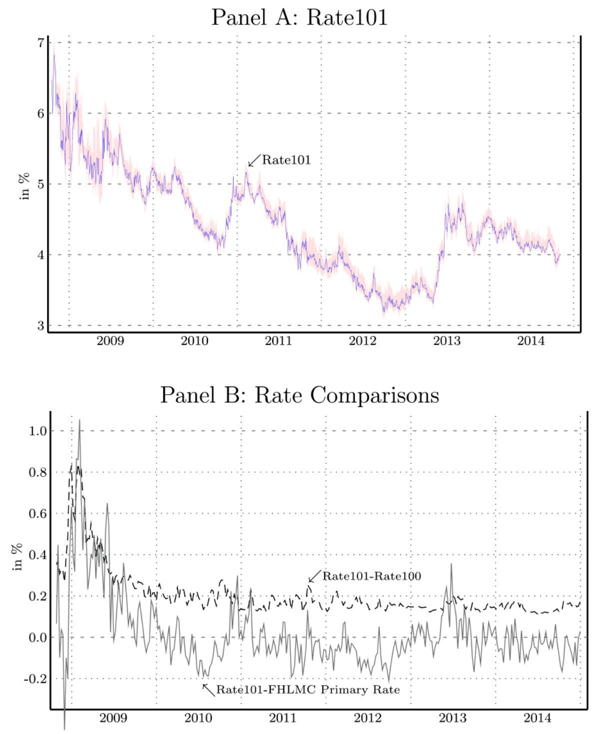


Figure 3. Mortgage rates over our sample period. The line in Panel A shows the median rate across lenders at which a rebate of +1 point is offered ("Rate101"). The shaded area is the 10th to 90th percentile range of Rate101 offers across lenders in our data. Panel B shows the difference

(b) Figure 3: Mortgage rates

5.3 Figure 4: Event studies of major monetary policy announcements and Table II: Calibration of pass-through in QE events

Read: Figure 4 examines six QE-related events and shows MBS prices, borrower rebates, pass-through, and application volume. Table II formalizes the event pass-through calculations.

Economic message. QE can raise MBS prices immediately, but the consumer mortgage benefit depends on intermediary capacity at the time of the shock.

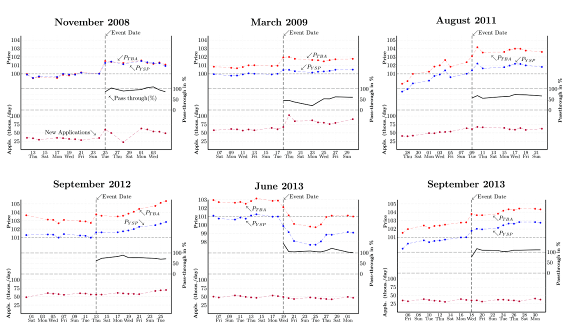


Figure 4. Event studies of major monetary policy (QE) announcements over our sample period. The top lines show MBS prices (p_{TBA}) and median rebates offered to borrowers (p_{YSP}), holding the rate fixed at Rate101 (Rate100 in 2008 to 2009) of the day prior to the announcement. The middle line shows the pass-through of the TBA price change to YSPs. The bottom line shows daily application counts as measured by HMDA (nonweekends/nonholidays only). In each of the panels, the price data shown are based only on lenders who are present in the data for the entire event window. Note that in the top left panel, November 27 was Thanksgiving, which explains the missing data point and the low applications on Friday, November 28. (Color figure can be viewed at wileyonlinelibrary.com)

Table II

Calculation of Pass-Through for the Announcement of QE3 on September 13, 2012

	$\bar{t} = 9/12/2012$	$t = 9/13/2012$	
$Rate101, \bar{t}$	3.47	3.47	
$p_{YSP}^{Rate101,\bar{t}}$	101	101.6	$\Delta p_{YSP} = 0.6$
$p_{TBA}^{Rate101,\bar{t}}$	102.7	103.8	$\Delta p_{TBA} = 1.1$

The figure runs the same calculation for subsequent dates and shows that p_{TBA} drifted down over the next few days. Meanwhile, p_{YSP} stayed more or less the same, so the effective pass-through was somewhat higher at longer horizons.

5.4 Figure 5: High-frequency pass-through and Table III: Regressions of Δp_{YSP} on Δp_{TBA} at daily frequency

Read: Average daily pass-through is 0.918; negative MBS price changes pass through roughly one-for-one, while favorable changes pass through less. High application volume reduces pass-through of positive TBA price changes.

Economic message. Daily evidence shows both strong capital-market linkage and an important wedge: intermediaries absorb part of favorable MBS price changes, especially when demand is high.

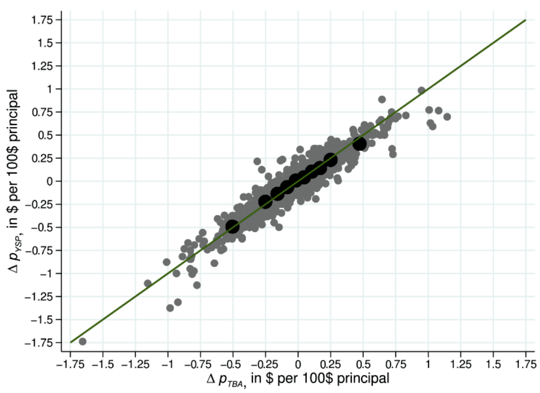


Figure 5. High-frequency pass-through. The figure provides a scatterplot of daily first dif-

(a) Figure 5: High-frequency pass-through

Table III

Regressions of Δp_{YSP} on Δp_{TBA} at Daily Frequency

This table presents regressions of daily changes in rebates (from lender rate sheets) on daily changes in MBS prices. Δp_{YSP} is the change from day $t-1$ to day t in the median rebate offered for a note rate corresponding to a rebate of 101 on day $t-1$. Δp_{TBA} similarly denotes the daily change in the price of the MBS for that same note rate (adjusted for guarantee fees) in the TBA market. Δp^{+}_{TBA} denotes the day-over-day change in TBA price for positive changes only, and Δp^{-}_{TBA} corresponds to negative changes only. Columns (3) and (5) also include a price change lagged by one business day. $Applications_{t-1}$ is the daily HMDA application volume from the previous business day, normalized to have a mean of zero and a standard deviation of one. Robust standard errors are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Dep. Var.: $\Delta p_{YSP,t}$	(1)	(2)	(3)	(4)	(5)
$\Delta p_{TBA,t}$	0.918***				
	(0.013)				
$\Delta p^{+}_{TBA,t}$		0.801***	0.795***	0.787***	0.782***
		(0.022)	(0.023)	(0.021)	(0.021)
$\Delta p^{-}_{TBA,t}$		1.022***	1.027***	1.025***	1.030***
		(0.022)	(0.022)	(0.022)	(0.022)
$\Delta p^{+}_{TBA,t-1}$			0.078***		0.088***
			(0.021)		(0.020)
$\Delta p^{-}_{TBA,t-1}$			0.014		0.007
			(0.023)		(0.023)
$\Delta p^{+}_{TBA,t} \times Applications_{t-1}$				−0.097***	−0.107***
				(0.022)	(0.023)
$\Delta p^{-}_{TBA,t} \times Applications_{t-1}$				0.010	0.020
				(0.022)	(0.022)
$\Delta p^{+}_{TBA,t-1} \times Applications_{t-1}$					−0.022
					(0.019)
$\Delta p^{-}_{TBA,t-1} \times Applications_{t-1}$					−0.028
					(0.022)
$Applications_{t-1}$				0.004	0.005
				(0.004)	(0.004)
Constant	−0.001	0.022***	0.016***	0.023***	0.015***
	(0.002)	(0.004)	(0.005)	(0.004)	(0.005)
Adj. R^2	0.88	0.89	0.89	0.89	0.89
Observations	1,431	1,431	1,431	1,431	1,431

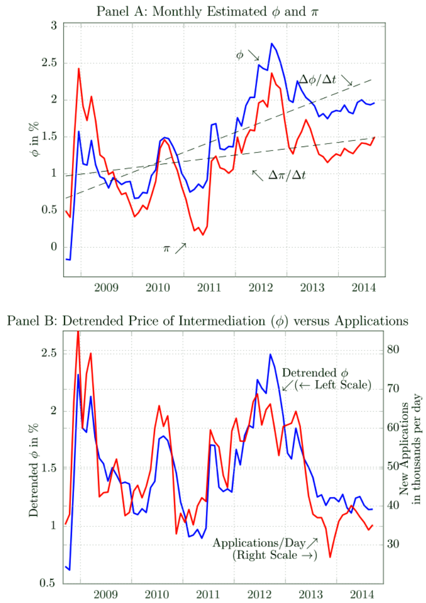
(and the number of originated loans stemming from these applications): a one-dollar change in p_{TBA} is on average associated with about a 7% change in applications and a 9% change in originations on the same day. The price changes

(b) Table III: Daily pass-through regressions

5.5 Figure 6: Evolution of price of intermediation over time and Table IV: Understanding time variation in the price of intermediation

Read: Figure 6 shows that ϕ and π are volatile and trend upward. Detrended ϕ and applications have correlation 0.78. Table IV shows that applications and time trend raise ϕ and π.

Economic message. Demand waves move ϕ in the short run, while postcrisis cost changes push the level upward over time.



re 6. Evolution of price of intermediation over time. In Panel A, ϕ denotes the mo
ıated average price of intermediation (calculated as the difference between the MBS n
of a loan with note rate Rate101 and the rebate offered to a borrower) across lend
sample, and π is the corresponding average gain-on-sale (which adjusts for the net
cing). The dashed line shows the estimated linear time trend in the ϕ series. Panel B
ıonthly average ϕ after detrending (and shifting the series so as to keep the average lev
as in Panel A), plotted against mortgage applications per business day (from HMDA

Figure 6: Evolution of price of intermediation

Table IV

Understanding Time Variation in the Price of Intermediation

This table presents regressions of the price of intermediation on nationwide mortgage applications, a linear time trend, and other variables. The unit of observation is the lender × month. ϕ denotes the monthly estimated average price of intermediation (calculated as the difference between the MBS market value of a loan with note rate Rate101 and the rebate offered to a borrower), and π is the corresponding average gain-on-sale (which adjusts for the net cost of servicing). All explanatory variables except for the time trend (where the unit is the calendar month) are standardized over the relevant sample, so that the coefficients can be interpreted as the effect of a one-standard-deviation change in the relevant variable on the price of intermediation. In the instrumental variables (IV) specifications, *Applications* are instrumented for by (WAC−10-year Treasury yield), as explained in the text (in Panel B, monthly changes in *Applications* are instrumented by Δ(WAC−10-year Treasury yield)). R^2 is not provided for IV specifications since it does not have a natural interpretation in this case. Standard errors (reported in parentheses) are clustered by lender and month. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Panel A: Levels Specification

Dep. Var.:	ϕ					π		$(\phi-\pi)$	
	OLS (1)	OLS (2)	OLS (3)	IV (4)	IV (5)	IV (6)	IV (7)	IV (8)	IV (9)
Applications	0.265***	0.328***	0.294***	0.358***	0.367***	0.404***	0.384***	−0.046*	−0.018
	(0.048)	(0.032)	(0.040)	(0.038)	(0.053)	(0.045)	(0.061)	(0.024)	(0.028)
Time Trend		0.027***	0.015**	0.028***	0.024***	0.015***	0.009	0.013***	0.015***
(monthly)		(0.002)	(0.006)	(0.002)	(0.006)	(0.003)	(0.008)	(0.002)	(0.005)
Interest Volat.			−0.075		−0.022		0.155**		−0.177***
			(0.053)		(0.058)		(0.061)		(0.033)
Lender Conc.			0.005		0.094		0.031		0.063
			(0.065)		(0.066)		(0.081)		(0.049)
R.E. Payroll			0.205***		0.162***		0.280***		−0.118***
			(0.048)		(0.053)		(0.060)		(0.031)
Lender FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Obs.	1,670	1,670	1,670	1,670	1,670	1,670	1,670	1,670	1,670
Adj. R^2	0.52	0.79	0.80						
Adj. R^2 (within)	0.20	0.64	0.67						
First-stage F-stat				207.7	74.3	207.7	74.3	207.7	74.3

(*Continued*)

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Table IV—*Continued*

Panel B: First-Difference Specification

Dep. Var.:	$\Delta\phi$				$\Delta\pi$		$\Delta(\phi-\pi)$	
	OLS (1)	OLS (2)	IV (3)	IV (4)	IV (5)	IV (6)	IV (7)	IV (8)
Δ *Applications*	0.191***	0.179***	0.274***	0.264***	0.348***	0.367***	−0.074***	−0.103***
	(0.044)	(0.053)	(0.051)	(0.050)	(0.063)	(0.056)	(0.028)	(0.024)
Δ *Int. Volat.*		0.030		0.090		0.207**		−0.117***
		(0.090)		(0.091)		(0.095)		(0.038)
Δ *Concentration*		−0.129*		−0.081		−0.056		−0.025
		(0.071)		(0.076)		(0.085)		(0.040)
Δ *R.E. Payroll*		0.014		0.006		0.038		−0.032
		(0.048)		(0.053)		(0.065)		(0.023)
Lender FE	Y	Y	Y	Y	Y	Y	Y	Y
Obs.	1,583	1,583	1,583	1,583	1,583	1,583	1,583	1,583
Adj. R^2	0.18	0.20						
Adj. R^2 (within)	0.19	0.20						
First-stage F-stat			72.9	63.2	72.9	63.2	72.9	63.2

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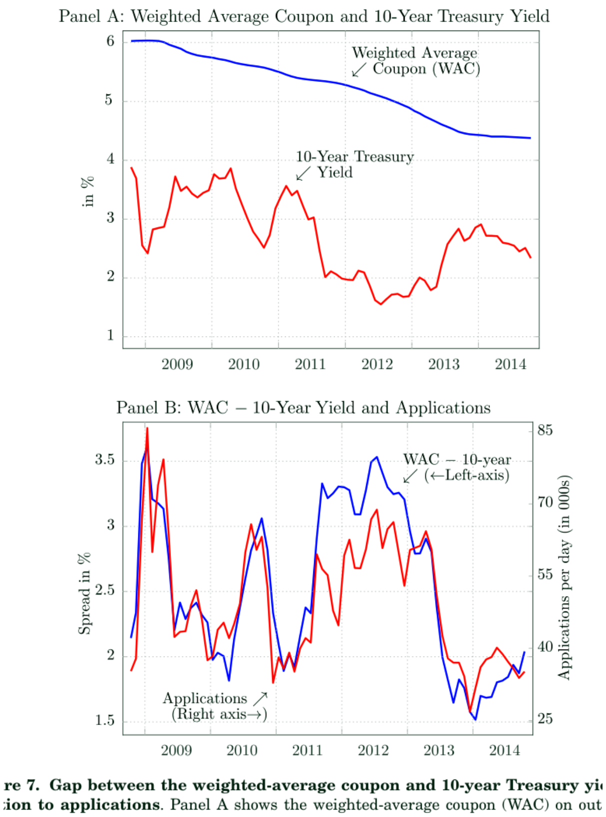
5.6 Figure 7: Gap between WAC and 10-year Treasury yield, Figure 8: Loan application volume and median processing time, and Table V: Processing-time regressions

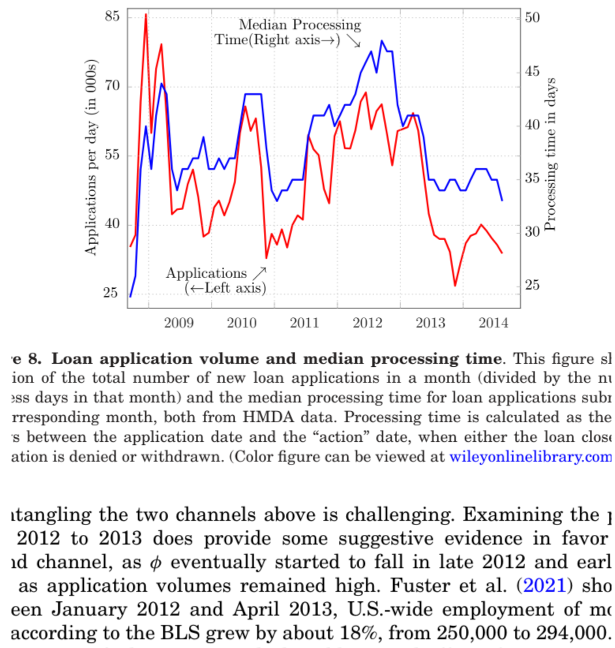
Read: Figure 7 captures refinancing incentives; Figure 8 shows applications and processing time moving together. Table V shows that a one-standard-deviation rise in applications raises processing time by about 3.1–3.5 days in OLS and 4.6–4.7 days in IV.

Economic message. Processing-time evidence directly supports capacity constraints.



re 7. Gap between the weighted-average coupon and 10-year Treasury yi
:ion to applications. Panel A shows the weighted-average coupon (WAC) on out

(a) Figure 7: WAC–Treasury gap



e 8. Loan application volume and median processing time. This figure sl
ion of the total number of new loan applications in a month (divided by the nu
ess days in that month) and the median processing time for loan applications subı
rresponding month, both from HMDA data. Processing time is calculated as the
s between the application date and the "action" date, when either the loan close
ation is denied or withdrawn. (Color figure can be viewed at wileyonlinelibrary.com

ıtangling the two channels above is challenging. Examining the p
2012 to 2013 does provide some suggestive evidence in favor
ıd channel, as ϕ eventually started to fall in late 2012 and earl
as application volumes remained high. Fuster et al. (2021) sho
een January 2012 and April 2013, U.S.-wide employment of mo
according to the BLS grew by about 18%, from 250,000 to 294,000.

(b) Figure 8: Applications and processing time

Table V

Regression of Processing Time on Volume of Loan Applications and Time Trend

This table presents results from loan-level regression using HMDA data from January 2008 to October 2014. The dependent variable, processing time, is defined as the number of days between loan application and action (accept, withdraw, or deny). The variable *Applications*, measured monthly, corresponds to the count of loan applications from HMDA divided by the number of business days in a month. We normalize this measure to have a mean of zero and a standard deviation of one across months in the sample. In columns (3) and (6), *Applications* are instrumented for by (WAC − 10-year Treasury yield). Loan controls in columns (2), (3), (5), and (6) include the log of applicant income, the log of the loan amount, and indicators for Federal Housing Administration loans, Veterans Affairs loans, jumbo loans, loan purpose (purchase or refi), occupancy, property type, applicant race, gender, whether the loan has a coapplicant, whether a preapproval was obtained, and whether income was missing from the application. In columns (1) to (3), we control for the action that was taken on the application. The sample includes applications for first-lien mortgages on single-family properties. R^2 is not provided for IV specifications since it does not have a natural interpretation in this case. Standard errors (reported in parentheses) are clustered by lender and month. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Dep. Var.: *Processing Time, in Days*	OLS (1)	OLS (2)	IV (3)	OLS (4)	OLS (5)	IV (6)
Applications	3.508***	3.088***	4.613***	3.155***	3.141***	4.731***
	(0.471)	(0.451)	(0.623)	(0.501)	(0.494)	(0.689)
Time Trend (monthly)	0.128***	0.0922***	0.118***	0.170***	0.183***	0.211***
	(0.0337)	(0.0289)	(0.0351)	(0.0273)	(0.0280)	(0.0350)
Sample	All	All	All	Originated	Originated	Originated
Census Tract FEs	No	Yes	Yes	No	Yes	Yes
Lender FEs	No	Yes	Yes	No	Yes	Yes
Loan Controls	No	Yes	Yes	No	Yes	Yes
Adj. R^2	0.05	0.16		0.16	0.17	
First-stage F-Stat			160.0			163.1
Observations	80,447,868	80,447,474	80,447,474	50,802,197	50,802,197	50,802,197

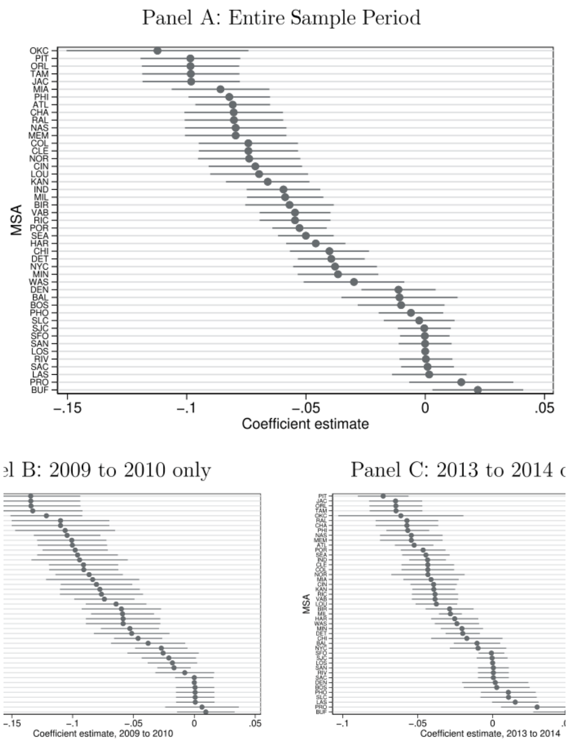
processing times can be substantial; for instance, it goes from 25 days to about

Table V: Processing-time regressions

5.7 Figure 9: Estimated differences in the price of intermediation across MSAs and Table VI: MSA-level regressions

Read: Figure 9 reports MSA fixed effects relative to Los Angeles. Table VI tests whether local foreclosure, forbearance, unemployment, judicial foreclosure law, concentration, and other variables explain these differences.

Economic message. The postcrisis rise in ϕ appears broad-based rather than concentrated in distressed local housing markets or more concentrated local lending markets.



Estimated differences in the price of intermediation across M
s estimated MSA fixed effects (with Los Angeles as the omitted base categor
95% confidence intervals based on regressions of the monthly price of inte
y-month fixed effects and MSA fixed effects. The price of intermediation is ca
he loan amount and under the assumption that p_{TBA} is identical across M
timates for the entire 2009 to 2014 period; Panel B for 2009 to 2010 only, a
2014 only. Standard errors underlying confidence intervals are clustered b

a MSA j in month $t-1$ (similar to Scharfstein and Sunders

(a) Figure 9: MSA differences

Table VI

Regressions of MSA-Level Price of Intermediation on Local Market Concentration and Default Rates

This table presents regressions of the price of intermediation on local market concentration and local mortgage default rates. The unit of observation is lender i × metropolitan statistical area (MSA) j × month t. ϕ denotes the monthly estimated average price of intermediation. *Top 4 Share* is the one-month-lagged market share of the largest four lenders in an MSA, following Scharfstein and Sunderam (2013). This variable is standardized to have a mean of zero and a standard deviation of one across all MSA-month observations. *Delinq. Rate* is the one-month-lagged 60+ day delinquency rate of loans originated in an MSA over the previous 24 months as measured in GSE public loan-level disclosure data, following Hurst et al. (2016). This variable is normalized to have a mean of zero and a standard deviation of one across MSAs within a given month. Standard errors (reported in parentheses) are clustered by lender and MSA. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Dep. Var.: ϕ_{ijt}	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Top 4 Share	−0.160***	0.002	−0.006*				0.000	−0.005*
	(0.023)	(0.006)	(0.003)				(0.005)	(0.003)
Delinq. Rate				−0.024*	−0.015***	−0.004**	−0.015***	−0.004*
				(0.014)	(0.004)	(0.002)	(0.004)	(0.002)
Lender FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	No	Yes	Yes	No	Yes	Yes	Yes	Yes
MSA FE	No	No	Yes	No	No	Yes	No	Yes
Mean(dep. var.)	1.53	1.53	1.53	1.53	1.53	1.53	1.53	1.53
St. Dev. (dep. var.)	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68
Obs.	136,231	136,231	136,231	136,231	136,231	136,231	136,231	136,231
Adj. R^2	0.42	0.78	0.78	0.37	0.78	0.78	0.78	0.78

Results of these regressions appear in Table VI. In a first specification, we only add lender fixed effects. Column (1) shows that in this specification, the relationship between ϕ and market concentration is strongly negative, in

(b) Table VI: MSA-level regressions

5.8 Table VII: Measuring trends in different components of origination costs of mortgage banks and Figure 10: Measuring the profits from intermediation

Read: Table VII shows total direct loan production expenses trending upward by 8.274 bps per quarter, with personnel expenses contributing 5.642 bps. Figure 10 shows that rising ϕ is primarily cost-related rather than a persistent increase in pure profits.

Economic message. Postcrisis mortgage production became more labor- and compliance-intensive.

Table VII

Measuring Trends in Different Components of Origination Costs of Mortgage Banks

This table presents regressions of loan production expenses, personnel expenses, and productivity measures on a linear time trend and loan origination volumes. The data source is Table B2 in the *Quarterly Mortgage Bankers Performance Report* published by the Mortgage Bankers Association. Panels A and B use the simple average across reporting firms (which is all that is available in the report); Panel C uses the weighted average (by number of loans originated). In Panel A, columns (2) to (5) add up to column (1); similarly, columns (1) to (5) of Panel B add up to column (2) of Panel A. Newey-West standard errors (three lags) are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Panel A: Total Direct Loan Production Expenses and Four Subcategories

(in basis points)	Total (1)	Personnel (2)	Occup.&Equipm. (3)	Tech-Rel. Exp. (4)	Other Exp. (5)
Time Trend	8.274***	5.642***	0.697***	0.203***	1.732***
(quarterly)	(0.645)	(0.408)	(0.085)	(0.037)	(0.192)
Avg. Loans Orig.	−0.378***	−0.237***	−0.050***	−0.009***	−0.082***
(USD mn)	(0.046)	(0.030)	(0.005)	(0.002)	(0.013)
Constant	−1,346.331***	−926.794***	−113.078***	−34.662***	−271.797***
	(122.532)	(77.925)	(16.374)	(7.147)	(36.708)
Avg. Cost (bp)	250.68	168.47	16.02	4.67	61.52
Adj. R2	0.89	0.90	0.79	0.71	0.83
Obs.	26	26	26	26	26

Panel B: Decomposing Personnel Expenses

(in basis points)	Sales (1)	Fulfillment (2)	Prod. Support (3)	Mgmt&Directors (4)	Benefits (5)
Time Trend	1.878***	1.528***	0.731***	0.485***	1.020***
(quarterly)	(0.128)	(0.164)	(0.094)	(0.072)	(0.106)
Avg. Loans Orig.	−0.058***	−0.066***	−0.047***	−0.013***	−0.053***
(USD mn)	(0.008)	(0.011)	(0.007)	(0.004)	(0.008)
Constant	−295.974***	−255.922***	−118.376***	−81.823***	−174.700***
	(25.309)	(31.281)	(17.442)	(13.843)	(19.714)
Avg. Cost (bp)	74.75	40.22	18.63	14.63	20.24
Adj. R^2	0.91	0.83	0.74	0.82	0.79
Obs.	26	26	26	26	26

Panel C: Productivity

	Monthly Number of Loan Closings per…			
	Total Prod. Empl. (1)	Sales Empl. (2)	Fulfillment Empl. (3)	Prod. Support Empl. (4)
Time Trend	−0.077***	−0.110***	−0.299***	−0.465***
(quarterly)	(0.009)	(0.009)	(0.047)	(0.067)
Avg. Loans Orig.	0.001***	0.004***	0.003***	0.008***
(#)	(0.000)	(0.000)	(0.001)	(0.001)
Constant	16.306***	22.623***	64.065***	99.204***
	(1.622)	(1.766)	(8.886)	(12.834)

(*Continued*)

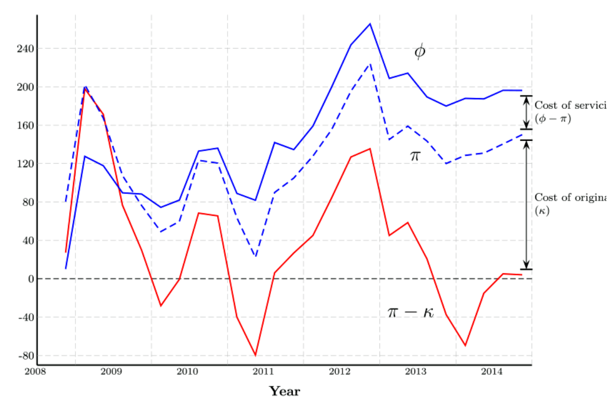
Table VII: Cost trends

Table VII—*Continued*

Panel C: Productivity

	Monthly Number of Loan Closings per…			
	Total Prod. Empl. (1)	Sales Empl. (2)	Fulfillment Empl. (3)	Prod. Support Empl. (4)
Avg. Y	2.36	4.86	6.77	13.45
Adj. R^2	0.83	0.92	0.78	0.72
Obs.	26	26	26	26

(a) Table VII continued



gure 10. Measuring the profits from intermediation. This figure shows time series of imated price of intermediation and its components. As explained in the text, ϕ is our broad asure of the price of intermediation. π (gain-on-sale) nets out the direct costs of servicing n. To construct κ, the direct cost of originating a loan, we use data from the Mortgage Bank sociation on all loan production costs less income received by the intermediary directly from

(b) Figure 10: Profits from intermediation

5.9 Table VIII: Spending on intermediation over our sample period

Read: Total spending on intermediation is estimated at \$155.0 billion. Origination/gain-on-sale accounts for \$131.7 billion and servicing for \$23.3 billion. The upward trend contributes \$97.4 billion.

Economic message. The wedge is economically large and changes the household cost of mortgage credit.

Table VIII

Spending on Intermediation over Our Sample Period in Billions of Dollars

This table presents estimates of total spending on intermediation over the period October 2008 to October 2014, and decomposes it into different components. Estimates use monthly values from Panel A of Figure 6 multiplied by origination amounts from HMDA data. Contribution of the time trend equals $\sum_{t=1}^{T} \hat{\beta}_{TimeTrend} \times (t-1) \times Originations(in\ \$)_t$ and contribution of applications equals $\sum_{t=1}^{T} \hat{\beta}_{Apps} \times Apps \times Originations(in\ \$)_t$, where the estimated $\hat{\beta}$'s for ϕ and π come from columns (4) and (6) of Table IV, respectively. Applications (*Apps*) are standardized in these regressions to have a mean of zero and a standard deviation of one.

	All Loans	Purchase	Refinance
Estimated Spending on Intermediation (10/2008–10/2014)			
Total (ϕ)	155.0	54.9	100.1
= Origination (π)	131.7	44.5	87.3
+ Servicing ($\phi - \pi$)	23.3	10.5	12.8
Estimated Contribution of Time Trend			
Total (ϕ)	97.4	38.5	58.9
= Origination (π)	52.2	20.7	31.6
+ Servicing ($\phi - \pi$)	45.2	17.9	27.3
Estimated Contribution of Dependence on Application Volume			
Total (ϕ)	11.6	−0.1	11.7

Table VIII: Aggregate spending

6 Short summary

- The paper develops a new measure of the mortgage intermediation price using rate sheets and TBA/MBS prices.
- Pass-through from MBS prices to borrower rebates is high but incomplete and weaker during high application demand.
- Monthly ϕ and π are volatile, demand-sensitive, and upward trending.
- Processing-time and cost evidence support capacity constraints and rising postcrisis production costs.
- Aggregate spending on intermediation is large: about \$155 billion over the sample, with about \$97 billion attributed to the upward trend.

7 Conclusion

7.1 Core conclusion

Fuster, Lo, and Willen show that the mortgage market's link between capital markets and households is mediated by a large and time-varying wedge. This price of intermediation rises with demand, responds to capacity constraints, and trends upward over the postcrisis period.

7.2 What the paper changes conceptually

The paper inserts an explicit intermediary supply curve between MBS prices and borrower mortgage terms. Monetary policy can move MBS prices, but borrower terms depend on how much intermediaries pass through through rebates and rate sheets.

7.3 Economic mechanism

Refinancing waves create spikes in demand for underwriting, compliance, document review, and pipeline management. Since staffing and systems cannot adjust instantly, lenders ration capacity partly through price and partly through longer processing times.

7.4 Policy implications

QE is most powerful for households when intermediary capacity is slack. During refinancing booms, part of policy-induced MBS price gains becomes intermediary margin rather than borrower benefit. Postcrisis regulation likely raised measured costs, although some of those costs may reflect valuable increases in origination and servicing quality.

7.5 Limitations and open questions

The paper cannot fully separate socially valuable compliance from deadweight cost, does not observe all third-party borrower fees, and focuses on the TPO rate-sheet channel. A remaining question is how intermediary capacity should be modeled dynamically.

7.6 Takeaway

Financial intermediation prices are state variables. They move with demand, capacity, regulation, and operational technology. Models of mortgage-market pass-through that assume a frictionless link from secondary-market yields to borrower terms miss a central part of the mortgage market.